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CMPG 325 : computer networks

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Project ID:CM PG 325-2026-145

Client background

Amohelang Electronics Repairs & Sales (Vryburg) is a technology-sector business located in Vryburg, North West Province, South Africa. The business was established to serve both urban and rural communities around Vryburg with affordable electronics sales and professional repair services.

Core Business Operations

The organization operates two main revenue streams. First, front-of-house retail sales of consumer electronics including cell phones, accessories, chargers, small appliances, and components. Second, a dedicated technical workshop offering diagnostic, repair, and maintenance services for phones, laptops, TVs, and other electronic devices. This dual operation requires reliable internal communication between departments while keeping sales and workshop traffic segmented for security and performance.

What the organisation requires :

1. A functioning internal network that supports day-to-day departmental operations.
2. Automatic IP address assignment.
3. Room to expand for future growth.
4. Successful data exchange between appropriate nodes.
5. A working, testable implementation in Cisco Packet Tracer.

How we aim to fix the networking problems at Amohelang Electronics Repairs and Sales

1. Design a multi-VLAN network using VLSM on 192.168.61.0/24.
2. Configure DHCP (scoped multi-VLAN address assignment) Foundational challenge. Each VLAN gets its own DHCP pool from the router.
3. Implement Router-on-a-Stick for inter-VLAN routing.
4. Create a shared printer zone that serves two departments that currently cannot print (CR8).
5. Reserve one VLAN and one subnet for a new department to be added next year .
6. Verify and demonstrate with ipconfig, ping, and traceroute that data exchange works and DHCP works.

Departmental Structure

Department	Function	Printer access today.
Sales	Front-of-house sales staff	No
Repairs / Workshop	Technicians and repair bench PCs	No
Admin & Management	Office, accounts, ordering	Yes (existing)
Future Department (reserved)	Not yet defined — reserved for growth	N/A

Design Constraint

Constraint: A new department will be added next year. The addressing plan must accommodate this without re-addressing the existing network. This is satisfied by reserving an unused VLAN and subnet (VLAN 40) for the future department from the outset.

Client Change Request CR8

A shared printer zone must serve two departments that currently cannot print. Sales and Repairs currently have no printer; Admin already has one. The design introduces a dedicated Printer VLAN (VLAN 50) that Sales and Repairs are routed to, satisfying CR8 exactly as needed without extending scope to departments that do not need it.

Testing Requirements

- Verify relevant end-to-end connectivity between departmental PCs and the printer zone.
- Verify the assigned DHCP (scoped multi-VLAN) challenge —,each VLAN receives addresses only from its own scope.
- Capture evidence of successful configuration and operation (screenshots, command output).
- Document any troubleshooting performed during implementation.